

Healthplus

use healthplus;

Data Profiling and Cleaning

Members data profiling

SELECT distinct gender from Members;

	gender
▶	Female
	Male

SELECT * from Members where email is null;

	member_id	first_name	last_name	gender	date_of_birth	age	city	membership_type	registration_date	phone_number	email
▶	M000025	Puneet	Agarwal	Male	1994-07-21	31	Kozhikode	Family	2025-06-12	8845664523	NULL
	M000050	Amit	Sharma	Male	1989-08-22	36	Gurugram	Corporate	2021-10-16	8054760064	NULL
	M000066	Tushar	Verma	Male	1970-06-18	55	Kochi	Corporate	2024-11-30	8058567855	NULL
	M000086	Priyal	Saxena	Female	1978-12-15	47	Vijayawada	Individual	2025-12-09	8103993823	NULL
	M000120	Riya	Chopra	Female	1983-04-10	42	Delhi	Corporate	2025-09-30	8974893725	NULL
	M000163	Riya	Sharma	Female	1978-12-14	47	Kozhikode	Individual	2023-08-12	8055916091	NULL
	M000176	Shweta	Malhotra	Female	1965-11-06	60	Mumbai	Corporate	2023-01-30	8043943221	NULL
	M000190	Vivek	Bansal	Male	1994-02-09	31	Mangaluru	Corporate	2022-07-07	8868829719	NULL
	M000214	Karan	Bhatia	Male	1956-04-28	69	Bengaluru	Corporate	2024-02-17	8396193734	NULL
	M000234	Tushar	Bansal	Male	1966-10-11	59	Chennai	Individual	2024-08-05	8019335856	NULL
	M000266	Rajat	Gupta	Male	1971-04-03	54	Mumbai	Corporate	2023-08-02	8719568930	NULL
	M000304	Bhavna	Kulkarni	Female	1963-03-05	62	Bengaluru	Individual	2024-10-13	8940991186	NULL
	M000341	Aryan	Pandey	Male	1968-05-14	57	Madurai	Family	2022-06-22	8338986836	NULL

SELECT * from Members where email not like '%@%.%';

	member_id	first_name	last_name	gender	date_of_birth	age	city	membership_type	registration_date	phone_number	email
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Members data cleaning

UPDATE Members

set gender=

CASE

WHEN lower(trim(gender)) in ('male','m')

THEN 'Male'

WHEN lower(trim(gender)) in ('female','f')

THEN 'Female'

ELSE gender

END;

set sql_safe_updates=0;

```

update Members
set email=
CASE
WHEN email like '%gmail' and email not like '%@%'
THEN REPLACE(email,'gmail','@gmail')
END
where email not like '%@%.%';

```

Claims data profiling

SELECT * from Claims where consultation_id is null;

	claim_id	member_id	consultation_id	claim_amount	claim_date	claim_status	insurance_provider
▶	CL00006	M000698	NULL	6186.17	2025-02-26	Approved	Bajaj Allianz General Insurance
	CL00012	M002137	NULL	4769.12	2025-03-05	Rejected	HDFC ERGO Health Insurance
	CL00016	M002386	NULL	6147.44	2024-02-26	Rejected	HDFC ERGO Health Insurance
	CL00026	M001389	NULL	8208.55	2025-04-20	Approved	Niva Bupa Health Insurance
	CL00027	M001226	NULL	6344.27	2024-03-23	Approved	Star Health Insurance
	CL00029	M000769	NULL	1309.64	2025-08-09	Settled	HDFC ERGO Health Insurance
	CL00039	M000240	NULL	3617.00	2025-11-08	Approved	ICICI Lombard General Insurance
	CL00040	M000161	NULL	11220.84	2024-06-10	Settled	HDFC ERGO Health Insurance
	CL00051	M000247	NULL	14180.04	2025-02-28	Approved	Niva Bupa Health Insurance
	CL00054	M002079	NULL	4639.09	2025-01-21	Settled	Care Health Insurance

SELECT * from Claims where insurance_provider is null;

	claim_id	member_id	consultation_id	claim_amount	claim_date	claim_status	insurance_provider
▶	CL00062	M000349	CN003650	6199.57	2024-06-18	Approved	NULL
	CL00091	M001252	CN000901	8286.64	2025-03-05	Approved	NULL
	CL00137	M000859	CN000482	7269.33	2025-02-27	Pending	NULL
	CL00144	M001606	NULL	6645.95	2025-05-29	Settled	NULL
	CL00178	M000643	CN001389	3575.81	2024-06-12	Approved	NULL
	CL00228	M002107	NULL	2588.08	2024-05-06	Pending	NULL
	CL00277	M000896	CN005502	7615.79	2024-06-23	Pending	NULL
	CL00315	M001453	CN004358	13674.99	2025-12-17	Settled	NULL
	CL00330	M000025	NULL	14508.63	2025-12-10	Pending	NULL
	CL00333	M001015	CN002027	10015.57	2024-12-02	Approved	NULL

Billing data profiling

SELECT * from Billing where consultation_id is null;

	bill_id	member_id	consultation_id	bill_date	consultation_charges	lab_charges	medicine_charges	total_amount	bill_status
▶	BI000005	M002346	NULL	2024-11-02	262.97	1034.57	333.77	1631.31	Partially Paid
	BI000019	M001472	NULL	2025-10-10	1129.61	467.76	211.63	1809.00	Partially Paid
	BI000026	M001907	NULL	2024-06-17	1031.88	782.36	124.23	1938.47	Paid
	BI000032	M000050	NULL	2025-12-24	838.95	2292.36	600.46	3731.77	Unpaid
	BI000050	M002388	NULL	2025-09-20	1069.92	2860.51	22.78	3953.21	Paid
	BI000058	M001308	NULL	2024-12-21	456.81	924.15	214.65	1595.61	Paid
	BI000059	M001478	NULL	2024-09-08	303.65	151.28	733.64	1188.57	Paid
	BI000067	M000252	NULL	2024-08-05	1009.20	1252.41	429.11	2690.72	Paid
	BI000071	M002444	NULL	2025-03-03	877.18	337.50	534.01	1748.69	Partially Paid
	BI000078	M001556	NULL	2024-07-29	353.13	1110.72	1465.97	2929.82	Paid
	BI000085	M001197	NULL	2025-07-19	472.00	1420.46	1206.93	3099.39	Unpaid

Payments data profiling

SELECT * FROM Payments where payment_mode is null;

	payment_id	bill_id	member_id	payment_date	payment_amount	payment_mode	payment_status
▶	PA000004	BI000004	M001254	2024-08-05	2389.26	NULL	Success
	PA000006	BI000006	M000411	2024-04-06	1425.81	NULL	Success
	PA000072	BI000072	M000493	2025-05-06	2298.61	NULL	Success
	PA000094	BI000094	M001662	2024-03-31	4730.57	NULL	Success
	PA000109	BI000109	M002437	2025-12-22	2801.17	NULL	Success
	PA000144	BI000144	M001236	2025-03-25	4850.14	NULL	Success
	PA000215	BI000215	M001725	2025-01-13	3411.15	NULL	Success
	PA000246	BI000246	M002302	2024-01-21	3042.88	NULL	Success
	PA000313	BI000313	M002169	2025-08-30	1813.43	NULL	Failed
	PA000365	BI000365	M000334	2025-08-04	5144.07	NULL	Success
	PA000367	BI000367	M000498	2024-02-03	-2773.36	NULL	Success

Business Analysis

1.which clinics have the highest consultation workload

```
SELECT c.clinic_id,c.clinic_name,c.clinic_type,COUNT(co.consultation_id) AS  
consultation_count  
FROM Clinics c  
LEFT JOIN Consultations co  
ON c.clinic_id = co.clinic_id  
GROUP BY c.clinic_id  
ORDER BY consultation_count DESC;
```

	clinic_id	clinic_name	clinic_type	consultation_count
▶	C001	HealthPlus Diagnostic Clinic - Kozhikode	Diagnostic	945
	C017	HealthPlus Diagnostic Clinic - Pune	Diagnostic	337
	C012	HealthPlus Primary Care Clinic - Mangaluru	Primary Care	317
	C004	HealthPlus Primary Care Clinic - Delhi	Primary Care	304
	C003	HealthPlus Telemedicine Hub Clinic - Hyderabad	Telemedicine Hub	303
	C015	HealthPlus Telemedicine Hub Clinic - Hyderabad	Telemedicine Hub	300
	C013	HealthPlus Diagnostic Clinic - Kozhikode	Diagnostic	298
	C011	HealthPlus Telemedicine Hub Clinic - Pune	Telemedicine Hub	297
	C007	HealthPlus Telemedicine Hub Clinic - Coimbatore	Telemedicine Hub	296
	C009	HealthPlus Diagnostic Clinic - Hyderabad	Diagnostic	296
	C018	HealthPlus Wellness Clinic - Kozhikode	Wellness	296
	C016	HealthPlus Primary Care Clinic - Madurai	Primary Care	295
	C002	HealthPlus Wellness Clinic - Kozhikode	Wellness	293
	C008	HealthPlus Primary Care Clinic - Vijayawada	Primary Care	290

2.Specialist Workload

```
SELECT s.specialist_id,s.first_name,s.specialization,COUNT(c.consultation_id) AS  
consultation_count  
FROM Specialists s  
LEFT JOIN Consultations c  
ON s.specialist_id = c.specialist_id  
GROUP BY s.specialist_id  
ORDER BY consultation_count DESC;
```

	specialist_id	first_name	specialization	consultation_count
►	SP0004	Charu	ENT Specialist	391
	SP0093	Sameer	Pulmonologist	70
	SP0080	Rajat	Physiotherapist	66
	SP0133	Sanya	Pediatrician	65
	SP0112	Nikhil	Pulmonologist	62
	SP0194	Sanya	Cardiologist	59
	SP0123	Anjali	Physiotherapist	55
	SP0088	Meghna	Dietitian	53
	SP0086	Diya	Diabetologist	52
	SP0118	Mansi	Diabetologist	48
	SP0155	Preeti	Nephrologist	48
	SP0173	Priyal	Dermatologist	48
	SP0055	Kirti	Dietitian	46
	SP0148	Arvind	Pulmonologist	46
	SP0220	Gaurav	General Physic...	46
	SP0085	Ishaan	Dentist	45
	SP0116	Sneha	Psychiatrist	45
	SP0074	Mansi	Diabetologist	44
	SP0082	Ritu	Gynecologist	44
	SP0021	Puneet	ENT Specialist	43
	SP0172	Mansi	Dietitian	43
	SP0217	Rajat	Nephrologist	43
	SP0109	Falquni	Nephrologist	42

3.Member Engagement and Repeat Visits

```
SELECT m.member_id,m.first_name,m.membership_type,COUNT(c.consultation_id) AS  
consultation_count  
FROM Members m  
LEFT JOIN Consultations c  
ON m.member_id = c.member_id  
GROUP BY m.member_id  
ORDER BY consultation_count DESC;
```

	member_id	first_name	membership_type	consultation_count
▶	M001338	Rohit	Individual	11
	M001441	Charu	Family	9
	M001362	Rajat	Family	9
	M000028	Dev	Family	8
	M002095	Mansi	Family	8
	M001917	Tanvi	Individual	8
	M000112	Ojasvi	Individual	8
	M000555	Charu	Family	8
	M002097	Rakesh	Family	7
	M000035	Shaurya	Corporate	7
	M000990	Aryan	Corporate	7

4. Telemedicine sessions by status

```
SELECT session_status,COUNT(*) AS session_count  
FROM Telemedicine_Sessions  
GROUP BY session_status  
ORDER BY session_count DESC;
```

	session_status	session_count
▶	Completed	1465
	Dropped	735

5. Telemedicine Condition and program status distribution

```
SELECT condition_name,program_status,COUNT(*) AS program_count
FROM Chronic_Care_Programs
GROUP BY condition_name, program_status
ORDER BY program_count DESC;
```

	condition_name	program_status	program_count
▶	Hypertension	Active	79
	Type 2 Diabetes	Active	70
	Arthritis	Active	68
	Asthma	Active	63
	Chronic Back Pain	Active	61
	Obesity Management	Active	60
	Chronic Kidney Disease	Active	54
	Thyroid Disorder	Active	53
	Cardiac Risk Management	Active	52
	Asthma	Completed	27
	Cardiac Risk Management	Completed	26
	Thyroid Disorder	Completed	26
	Chronic Kidney Disease	Completed	25

6.Specialist chronic-care workload

```
SELECT s.specialist_id,CONCAT(s.first_name, ' ', s.last_name) AS
specialist_name,s.specialization,COUNT(cc.program_id) AS chronic_program_count
FROM Specialists s
INNER JOIN Chronic_Care_Programs cc
ON s.specialist_id = cc.specialist_id
GROUP BY s.specialist_id,s.first_name,s.last_name,s.specialization
HAVING COUNT(cc.program_id) >= 5
ORDER BY chronic_program_count DESC;
```

	specialist_id	specialist_name	specialization	chronic_program_count
▶	SP0028	Kabir Ahluwalia	Psychiatrist	10
	SP0195	Manish Menon	Orthopedic	10
	SP0056	Nikhil Chawla	Physiotherapist	9
	SP0124	Diya Tiwari	Pulmonologist	9
	SP0103	Karan Rao	Physiotherapist	9
	SP0105	Puneet Iyer	Orthopedic	9
	SP0217	Rajat Malhotra	Nephrologist	9
	SP0062	Kavita Chopra	Gynecologist	8
	SP0166	Nidhi Malhotra	Orthopedic	8
	SP0031	Arvind Malhotra	Physiotherapist	7

7. Health Package-level subscription volume

```
SELECT
hp.package_id, hp.package_name, hp.package_type, hp.price, hp.validity_days, COUNT(ps.subscription_id) AS subscription_count
FROM Health_Packages hp
LEFT JOIN Package_Subscriptions ps
ON hp.package_id = ps.package_id
GROUP BY hp.package_id, hp.package_name, hp.package_type, hp.price, hp.validity_days
ORDER BY subscription_count DESC;
```

	package_id	package_name	package_type	price	validity_days	subscription_count
▶	PK002	Full Body Checkup Premium v2	Premium	8634.00	365	92
	PK023	Cardiac Risk Package v23	Cardiac	5986.00	180	89
	PK006	Full Body Checkup Premium v6	Premium	8430.00	365	86
	PK019	Senior Citizen Care Package v19	Senior Citizen	3826.00	365	83
	PK025	Senior Citizen Care Package v25	Senior Citizen	3834.00	365	82
	PK003	Senior Citizen Care Package v3	Senior Citizen	3532.00	365	81
	PK008	Diabetes Care Package v8	Diabetes Care	3159.00	180	81
	PK009	Basic Health Checkup v9	Basic	1689.00	365	81
	PK017	Basic Health Checkup v17	Basic	1708.00	365	80
	PK005	Cardiac Risk Package v5	Cardiac	5949.00	180	80
	PK029	Women's Wellness Package v29	Wellness	3092.00	365	80

8. Corporate enrollment and penetration

```
SELECT
co.corporate_id, co.company_name, co.industry, co.employee_count, COUNT(cm.corporate_member_id) AS enrolled_members
FROM Corporates co
LEFT JOIN Corporate_Members cm
ON co.corporate_id = cm.corporate_id
GROUP BY co.corporate_id, co.company_name, co.industry, co.employee_count
ORDER BY enrolled_members DESC;
```

	corporate_id	company_name	industry	employee_count	enrolled_members
▶	CO036	GlobalSoft Pvt Ltd	Education	1000	18
	CO044	DeltaSolutions Pvt Ltd	Consulting	4551	18
	CO042	ZenSolutions Pvt Ltd	Consulting	4991	18
	CO019	TechSystems Pvt Ltd	Telecommunications	2549	16
	CO058	NovaSolutions Pvt Ltd	Telecommunications	581	16
	CO027	SkyIndustries Pvt Ltd	Retail	2418	15
	CO007	SkyIndustries Pvt Ltd	E-commerce	2407	13
	CO010	NovaSolutions Pvt Ltd	Retail	4279	13
	CO026	SkyIndustries Pvt Ltd	Education	2950	13
	CO017	DeltaSystems Pvt Ltd	Pharmaceuticals	1174	13
	CO020	ApexLogistics Pvt Ltd	Telecommunications	1894	13
	CO008	TechSystems Pvt Ltd	Banking & Finance	3811	12

9. Most frequently prescribed medicines

```
SELECT medicine_name,COUNT(*) AS prescription_count
FROM Prescriptions
GROUP BY medicine_name
HAVING COUNT(*) >= 50
ORDER BY prescription_count DESC;
```

	medicine_name	prescription_count
▶	Paracetamol 650mg	554
	Vitamin D3	528
	Cetirizine 10mg	505
	Amlodipine 5mg	501
	Sertraline 50mg	497
	Metformin 500mg	496
	Montelukast 10mg	489
	Losartan 50mg	481
	Atorvastatin 10mg	475
	Pantoprazole 40mg	474

10. Diagnostic cost by test type

```
SELECT test_name,COUNT(*) AS test_count,SUM(test_cost) AS total_test_cost
FROM Lab_Tests
GROUP BY test_name
ORDER BY total_test_cost DESC;
```

	test_name	test_count	total_test_cost
▶	Liver Function Test	340	836750.77
	Kidney Function Test	342	832035.98
	Complete Blood Count	352	813212.53
	Chest X-Ray	324	786687.56
	Urine Routine	324	754800.96
	ECG	320	751054.54
	HbA1c	300	718936.38
	Vitamin D Test	303	701759.72
	Lipid Profile	298	691751.60
	Thyroid Profile	307	689588.49
	Vitamin B12 Test	290	648863.62

11. Insurance Claim status distribution

```
SELECT claim_status,COUNT(*) AS claim_count,SUM(claim_amount) AS  
total_claim_amount  
FROM Claims  
GROUP BY claim_status  
ORDER BY claim_count DESC;
```

	claim_status	claim_count	total_claim_amount
▶	Approved	616	4983465.96
	Settled	547	4306612.62
	Pending	471	3574079.49
	Rejected	166	1377894.19

12. Clinic satisfaction

```
SELECT cl.clinic_id,cl.clinic_name,COUNT(f.feedback_id) AS  
feedback_count,ROUND(AVG(f.rating), 2) AS average_rating  
FROM Clinics cl  
INNER JOIN Consultations c  
ON cl.clinic_id = c.clinic_id  
INNER JOIN Feedback f  
ON c.consultation_id = f.consultation_id  
GROUP BY cl.clinic_id, cl.clinic_name  
ORDER BY average_rating DESC;
```

	clinic_id	clinic_name	feedback_count	average_rating
▶	C015	HealthPlus Telemedicine Hub Clinic - Hyderabad	114	4.00
	C004	HealthPlus Primary Care Clinic - Delhi	113	3.98
	C006	HealthPlus Wellness Clinic - Coimbatore	117	3.93
	C016	HealthPlus Primary Care Clinic - Madurai	118	3.90
	C007	HealthPlus Telemedicine Hub Clinic - Coimbatore	135	3.87
	C014	HealthPlus Wellness Clinic - Coimbatore	119	3.86
	C008	HealthPlus Primary Care Clinic - Vijayawada	112	3.83
	C001	HealthPlus Diagnostic Clinic - Kozhikode	396	3.81
	C018	HealthPlus Wellness Clinic - Kozhikode	136	3.81
	C012	HealthPlus Primary Care Clinic - Mangaluru	121	3.80
	C010	HealthPlus Wellness Clinic - Mysuru	98	3.76
	C003	HealthPlus Telemedicine Hub Clinic - Hyderabad	130	3.73
	C017	HealthPlus Diagnostic Clinic - Pune	144	3.73

13. Staff distribution by clinic

```
SELECT c.clinic_id,c.clinic_name,s.designation,s.employment_type,COUNT(s.staff_id) AS  
staff_count,ROUND(AVG(s.salary), 2) AS average_salary  
FROM Clinics c  
LEFT JOIN Staff s  
ON c.clinic_id = s.clinic_id  
GROUP BY c.clinic_id, c.clinic_name,s.designation,s.employment_type  
ORDER BY c.clinic_id;
```

	clinic_id	clinic_name	designation	employment_type	staff_count	average_salary
▶	C001	HealthPlus Diagnostic Clinic - Kozhikode	Administrator	Full-Time	2	47632.20
	C001	HealthPlus Diagnostic Clinic - Kozhikode	Care Coordinator	Full-Time	1	57212.64
	C001	HealthPlus Diagnostic Clinic - Kozhikode	Front Desk Executive	Contract	1	55100.65
	C001	HealthPlus Diagnostic Clinic - Kozhikode	Front Desk Executive	Full-Time	6	57815.46
	C001	HealthPlus Diagnostic Clinic - Kozhikode	IT Support	Full-Time	3	55411.12
	C001	HealthPlus Diagnostic Clinic - Kozhikode	Lab Technician	Full-Time	3	53828.19
	C001	HealthPlus Diagnostic Clinic - Kozhikode	Nurse	Contract	1	88090.28
	C001	HealthPlus Diagnostic Clinic - Kozhikode	Nurse	Full-Time	4	44773.69
	C001	HealthPlus Diagnostic Clinic - Kozhikode	Pharmacist	Full-Time	5	66141.15
	C001	HealthPlus Diagnostic Clinic - Kozhikode	Pharmacist	Part-Time	1	66005.95
	C002	HealthPlus Wellness Clinic - Kozhikode	Administrator	Full-Time	3	47476.68
	C002	HealthPlus Wellness Clinic - Kozhikode	Care Coordinator	Full-Time	5	61811.83
	C002	HealthPlus Wellness Clinic - Kozhikode	Front Desk Executive	Full-Time	4	52944.49

KPI Analysis

1. Total Members

```
SELECT COUNT(*) AS total_members  
FROM Members;
```

	total_members
▶	2500

2. Total Specialists

```
SELECT COUNT(*) AS total_specialists  
FROM Specialists;
```

	total_specialists
▶	220

3. Total Clinics

```
SELECT COUNT(*) AS total_clinics  
FROM Clinics;
```

	total_clinics
▶	18

4. Total Consultations

```
SELECT COUNT(*) AS total_consultations  
FROM Consultations;
```

	total_consultations
▶	6000

5. Highest Consultation Fee

```
SELECT MAX(consultation_fee) AS highest_consultation_fee  
FROM Specialists;
```

	highest_consultation_fee
▶	1198.17

6. Total Telemedicine Sessions

```
SELECT COUNT(*) AS total_telemedicine_sessions  
FROM Telemedicine_Sessions;
```

	total_telemedicine_sessions
▶	2200

7. Total Package Subscriptions

```
SELECT COUNT(*) AS total_package_subscriptions  
FROM Package_Subscriptions;
```

	total_package_subscriptions
▶	3000

8. Active Package Subscriptions

```
SELECT COUNT(*) AS active_package_subscriptions  
FROM Package_Subscriptions  
WHERE expiry_date >= CURDATE();
```

	active_package_subscriptions
▶	273

9. Expired Package Subscriptions

```
SELECT COUNT(*) AS expired_package_subscriptions  
FROM Package_Subscriptions  
WHERE expiry_date < CURDATE();
```

	expired_package_subscriptions
▶	2727

10. Total Prescriptions

```
SELECT COUNT(*) AS total_prescriptions  
FROM Prescriptions;
```

	total_prescriptions
▶	5000

11. Total Laboratory Tests

```
SELECT COUNT(*) AS total_lab_tests  
FROM Lab_Tests;
```

	total_lab_tests
▶	3500

12. Total Laboratory Cost

```
SELECT ROUND(SUM(test_cost), 2) AS total_lab_cost  
FROM Lab_Tests;
```

	total_lab_cost
▶	8225442.15

13. Total Insurance Claims

```
SELECT COUNT(*) AS total_claims  
FROM Claims;
```

	total_claims
▶	1800

14. Total Insurance Claim Amount

```
SELECT ROUND(SUM(claim_amount), 2) AS total_claim_amount  
FROM Claims;
```

	total_claim_amount
▶	14242052.26

15. Total Billed Amount

```
SELECT ROUND(SUM(total_amount), 2) AS total_billed_amount  
FROM Billing;
```

	total_billed_amount
▶	18690854.89

16. Total Corporate Members

```
SELECT COUNT(DISTINCT member_id) AS corporate_members  
FROM Corporate_Members;
```

	corporate_members
▶	615